



Sapphire

△ **Rough sapphire** showing colour gradation

Both ruby and sapphire are gem varieties of the same mineral, corundum, an aluminium oxide that is next to diamond in hardness. Although commonly thought of as blue, sapphire can also be colourless, green, yellow, orange, violet, and pink, among other hues. Before the end of the 19th century, when geologists realized that sapphires of all colours were the same mineral, terminology regarding the naming of the gem persisted from medieval times: green sapphire was called Oriental peridot and yellow sapphire was Oriental topaz. One of the oldest known stones clearly identified as sapphire is St Edward's Sapphire: it is believed to date from the Anglo-Saxon king Edward the Confessor's coronet in 1042.

Fancy sapphires

With three exceptions, modern terminology simply uses the word "sapphire" preceded by the colour of the stone – for instance, yellow sapphire or green sapphire. Two exceptions are the rare pink-orange stones that are called padparadscha (Sanskrit for "lotus blossom"), and sapphire that appears blue in daylight and reddish or violet in artificial light, which is called alexandrine or alexandrite sapphire. The third exception is blue sapphire, which is simply called "sapphire". Colours other than blue are often referred to as fancy sapphires. Many sapphires, whatever their colour, have microscopic inclusions of rutile that produce a star when cut *en cabochon*.

Key pieces



Russian pectoral cross | Made in the Kremlin workshops in Moscow, Russia, during the second half of the 16th century, this cross was designed to be worn as a chest ornament. The central sapphire has been carved in the shape of Christ on the cross.



Logan sapphire | This flawless stone is the second largest known sapphire. The large table of the cushion cut shows the naturally perfect interior of this gem, which is set in a brooch surrounded by 20 round brilliant-cut diamonds.

Specification

Chemical name Aluminium oxide | **Formula** Al_2O_3
Colours Most colours | **Structure** Hexagonal, trigonal
Hardness 9 | **SG** 4.0–4.1 | **RI** 1.76–1.77 | **Lustre** Adamantine to vitreous | **Streak** Colourless



Round brilliant



Oval brilliant



Cameo



Step cut



Slab



Cabochon



Locations

1 Montana, North Carolina, USA **2** Colombia
3 Brazil **4** Kenya **5** Malawi **6** Sri Lanka **7** India
8 Kashmir **9** Thailand **10** Vietnam **11** Australia



Cartier clip brooch | The Duchess of Windsor revived the popularity of sapphires in the 1950s when she wore this clip. The sapphire is set in white gold and platinum, and smaller sapphire cabochons are used for the panther's spots.